PROJECT REPORT

Bank of Maharashtra Loan Product Assistant

ARAG chatassistant thatanswersloanquestions fromthe bank's official pages, shows its sources, and refuses when it cannot answer

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Purpose	Coursework walkthrough on building a RAG application
Source code	github.com/Shantanu-0101/BoM-Loan-Assistant

A learning project, not affiliated with Bank of Maharashtra. It is not financial advice.
Always confirm rates and terms with the bank.

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1. Business Understanding

1.1 The problem

Bank of Maharashtra publishes its loan information across many separate web pages and PDFs. This project's knowledge base alone covers 86 web pages and 3 PDF documents, holding 72,441 words and 115 tables. The details people care about most, such as rates, tenures, margins and fees, usually sit inside those tables, and they often depend on a condition such as the borrower's category, CIBIL score or where the salary account is held.

So a simple question can take real effort. "What is the maximum tenure for a personal loan if my salary account is with the bank?" means finding the right scheme, opening its page and reading the right row of a table. A general chatbot can reply faster, but it may answer from memory, give no source and state a wrong figure with confidence. In lending, a wrong figure can lead to a wrong decision.

1.2 What the project does about it

Theassistant letsa person ask inplain Englishand get a short answer built only from the bank's own pages. Each factual sentence carries a numbered citation that links to its source page. If the question is off-topic or the pages do not support an answer, the assistant says so instead of guessing.

Pain point	How the assistant helps
Information is spread over about 90 pages and PDFs	One question box searches all of it in a second or two
Key figures are buried in long tables	Table rows are kept whole and long tables repeat their header, so a figure stays with its conditions
Chatbot answers cannot be checked	Every factual sentence cites a page, and the reader can click through
Models can invent numbers	A code check removes any figure that is not in the retrieved text
Rates go out of date	Every answer states the date of its information
Off-topic or other-bank questions	A polite refusal, with no model call

1.3 Who it is for

Audience	What they get
Loan seekers	Quick answers on eligibility, tenure, margin, fees and documents before visiting a branch
Branch and support staff	A fast way to look up a scheme's terms, with the source page one click away
Students and developers	A complete, working RAG project with honest measurements to learn from

These are the intended audiences. The project is a learning project. It is not a service run by the bank and it has not been deployed to customers.

1.4 Scope

In scope	Out of scope
Home, personal, vehicle, education, gold, property, MSME and agriculture loans, plus retail rates and service charges, as published on bankofmaharashtra.bank.in	Other banks; accounts and other non-loan products; personal financial advice; applying for a loan; live rates (the data is a snapshot from 28 October 2026)

1.5 How success is measured

The project set onecleartarget:atleast80% of answers correct. It was tested on 25 questions written by hand from the official pages.

Measure	Result with hybrid search
Answers graded correct and complete by a person (target: at least 80%)	**24/25 (96%)**
Correct page found among the 5 passages sent to the model	22/22 (100%)
Out-of-scope questions refused	3/3
Real loan questions wrongly refused	0

Section 5 gives the full results, including what did not work and why 25 questions is a small sample.

2. Project Understanding

2.1 The core idea: RAG

RAGstands for **Retrieval-Augmented Generation**. Instead of asking a language model to remember the answer, the system first **finds** the most relevant passages, then asks the model to answer **from those passages only**. The model writes the sentences, but the facts come from the documents.

2.2 How it works

Thesystem has two phases. The **build phase** runs once and prepares the data:

- **Scrape.** Fetch the official pages and PDFs politely: obey robots.txt, wait 1 second between requests, cache everything and record it in a manifest.
- **Clean.** Keep the real text and tables, and drop menus, footers, buttons and scripts. This removed 74% of the words visible on the raw web pages.
- **Chunk.** Cut each page into 415 small pieces by its headings. A table row is not cut in half unless one row alone is over the 450-word limit, and every piece starts with its product name.
- **Index.** Turn each chunk into a list of 384 numbers that captures its meaning (an embedding) and store them in a FAISS index. A BM25 keyword index is also built from the same text.

The **answer phase** runs for every question:

- **Guard.** Refuse straight away if the question names another bank.
- **Search.** Take the top 20 chunks by meaning (FAISS) and the top 20 by exact words (BM25), then merge them with Reciprocal Rank Fusion and keep the best 5.
- **Guard again.** Refuse if even the best chunk is not similar enough (cosine similarity below 0.68).
- **Generate.** Send the 5 numbered chunks to the language model with strict rules: use only these passages, do no arithmetic, cite every factual sentence.
- **Check the numbers.** Remove any figure that does not appear in the chunks or the question.
- **Show.** Display the answer with linked citations, the source pages and the date of the information.

2.3 Files in the deliverables folder

File	What it is	How to use it
bom_loan_assistant_walkthrough.ipynb	The teaching notebook: 15 parts, with an explanation before every code cell, saved with real outputs	Open in Jupyter or VS Code and go cell by cell
bom_loan_assistant_walkthrough.py	The same content as the notebook, cell for cell, as a script	python deliverables/ bom_loan_assistant_walkthrough.py
streamlit_app.py	The chat app, commented section by section	streamlit run deliverables/streamlit_app.py
BoM_Loan_Assistant_Project_Report.pdf	This report	Read or share

All of these run from inside the project folder, because they read its data/, index/ and src/ folders. The notebook also re-writes the core logic in readable cells, then checks itself against the real package.

2.4 Tools and libraries

Job	Tool	Why this one
Language	Python 3.13	Built and tested on this; needs 3.12 or newer
Scraping	requests, BeautifulSoup, lxml	Pages are plain HTML, so no browser automation is needed
PDF text	PyMuPDF	Reads PDF text; scanned PDFs have none and are left out
Embeddings	FastEmbed: BAAI/bge-small-en-v1.5	Runs locally on the CPU, free, no API quota
Vector search	FAISS (IndexFlatIP)	With 415 chunks, exact search is already instant
Keyword search	rank-bm25	Finds exact terms that meaning can miss, such as "LAD"
Answer model	Groq: openai/gpt-oss-120b	Free tier and fast
Fallback and judge	Google Gemini	Backup if Groq fails; second opinion in evaluation
User interface	Streamlit	One Python file gives a full chat app
Settings	pydantic-settings, .env	One place for settings; keys stay out of the code
Quality	pytest, ruff, mypy	135 unit tests, linting and type checks
Development help	Claude Code	Wrote most of the code and tests under the owner's direction, and helped prepare these course files

2.5 External services and APIs

Service	Used for	Settings	Used when
Groq chat completions	Writing answers	temperature 0, 2,048 output tokens at most, 30 s timeout	Every answered question
Gemini generate content	Fallback and evaluation judge	temperature 0, 30 s timeout	Only if Groq fails, and during evaluation
bankofmaharashtra.bank.in	Source of all data	1 s between requests, robots.txt obeyed	Only when rebuilding the data

Embeddings, FAISS and BM25 all run on the local machine. Refused questions never reach a paid or rate-limited service. The free Groq tier allows about 8,000 tokens a minute and one answer costs about 2,500, so several questions asked quickly can hit the limit.

2.6 Key settings

Setting	Value	Plain meaning
candidates_k	20	Each search method brings back its 20 best chunks
top_k	5	The best 5 merged chunks go to the model
rrf_k	60	Smoothing constant in Reciprocal Rank Fusion
min_score	0.68	Refuse below this. In 24 test questions, real ones scored 0.74 or more and off-topic ones 0.66 or less
chunk_words / max	300 / 450	Target and maximum chunk size in words
chunk_overlap_words	50	Prose repeated between neighbouring chunks, so nothing is lost at a cut

3. Architecture Diagram

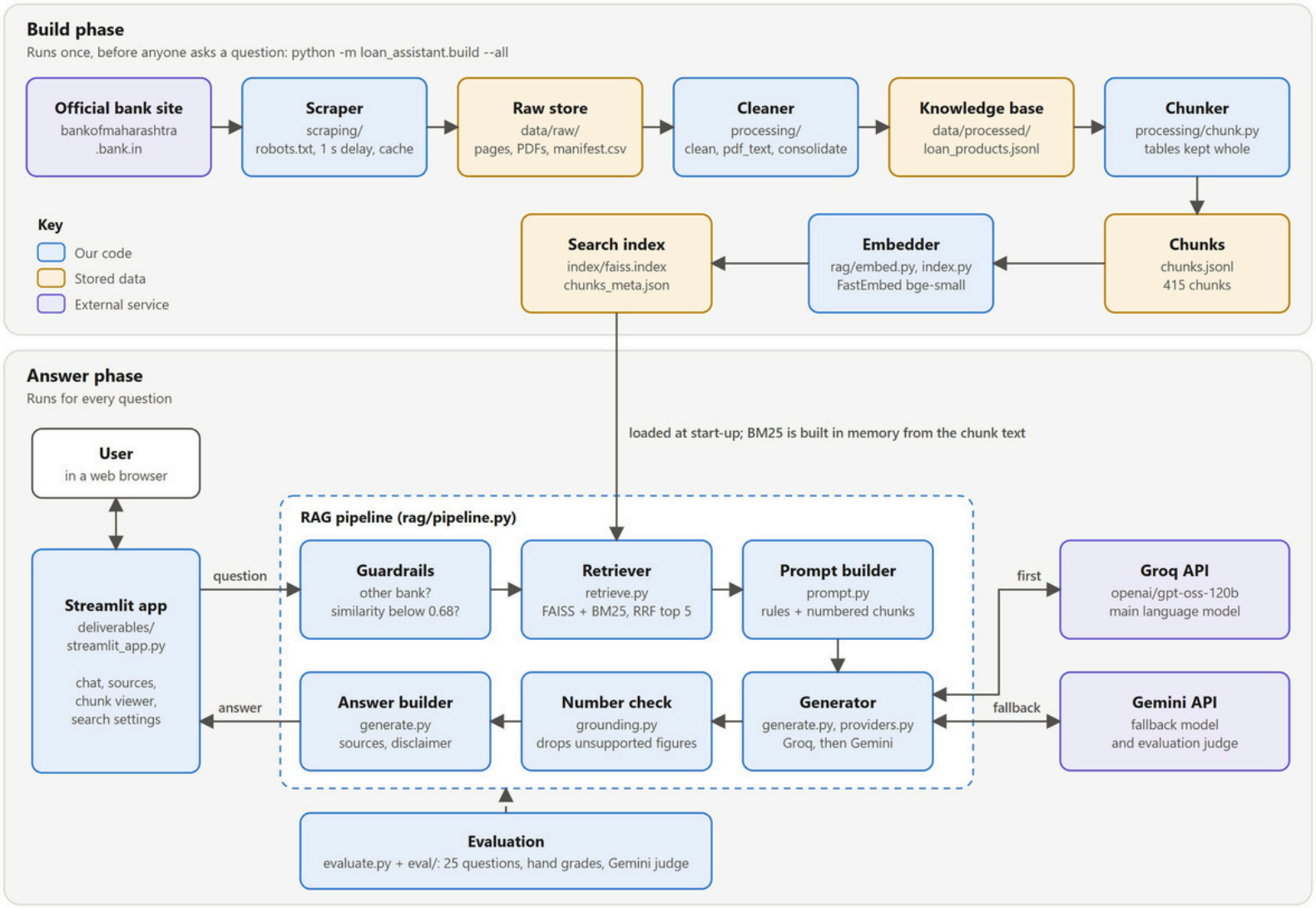


Figure 1. System architecture. File paths are inside src/loan_assistant/ unless they start with data/, index/, eval/ or deliverables/.

The system is split into two phases that share the search index. The build phase is slow and runs once. The answer phase is fast and runs for every question. The table below says what each box in Figure 1 does and where to find its code.

Component guide

Component	File	Responsibility
Scraper	scraping/discover.py, fetch.py, pdf.py	Find and fetch the loan pages politely, keep a manifest
Cleaner	processing/clean.py, pdf_text.py, consolidate.py	Turn raw pages into one clean knowledge base
Chunker	processing/chunk.py	Make chunks that respect sections and tables
Embedder	rag/embed.py, rag/index.py	Create the vectors and save the FAISS index
Retriever	rag/retrieve.py	Run FAISS and BM25 and merge them with RRF
Guardrails	rag/pipeline.py	Refuse other-bank and low-similarity questions
Prompt builder	rag/prompt.py	Hold the rules and number the context blocks
Generator	rag/generate.py, llm/providers.py	Call Groq or Gemini, tidy the output, list sources
Number check	rag/grounding.py	Remove figures that are not in the retrieved text
Streamlit app	deliverables/streamlit_app.py	Chat, sources, chunk viewer, search-mode switch
Evaluation	evaluate.py, eval/	Score both search modes on 25 questions

4. Flowchart

4.1Buildingtheknowledge base

Thesesixstepsrunonce.Thenumbersarefrom the real run on 28 October 2026.

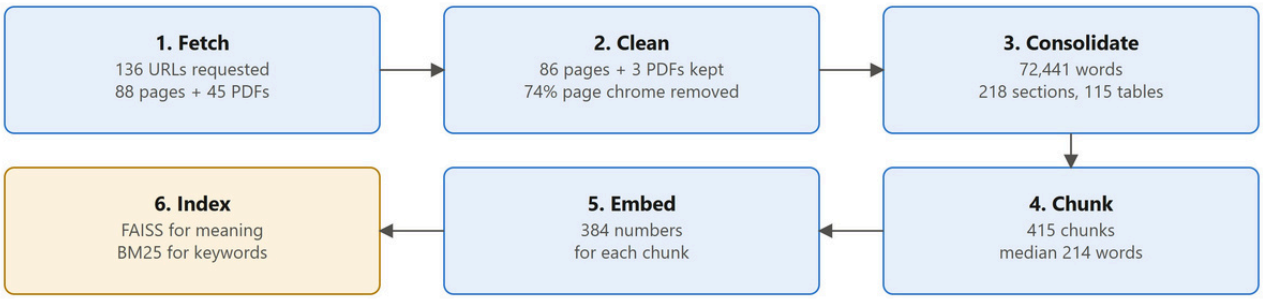


Figure 2. The build phase, with real counts at each step.

4.2 Answering a question

Thesestepsrun forevery question.Figure 3 on the next page shows every decision between the question and the answer. There are three separate chances to refuse, and only the path through the model uses API quota.

Step	What happens	Why
Other-bank check	A pattern looks for other banks' names. A question that also names Bank of Maharashtra is let through	The retriever would happily find home-loan text for "SBI home loan rate"
Search both ways	FAISS ranks by meaning and BM25 by exact words, 20 chunks each	Meaning misses exact terms, and keywords miss paraphrases
Merge with RRF	Each list gives a chunk 1 / (60 + rank), and the scores add up	The two kinds of score cannot be compared directly, but their ranks can
Similarity check	Refuse if the best cosine is below 0.68	Cheaper and safer than asking a model whether it knows
Build the prompt	Six rules, plus the 5 numbered chunks and the question	Each rule answers a problem seen in testing
Call the model	Groq first, then Gemini if it fails or returns nothing	One provider being down should not stop the app
Number check	Each figure must appear in the chunks or the question	The model once wrote 7.20% by subtracting 0.05 from 7.25, despite a rule against arithmetic
Finish	Add the disclaimer and list the cited pages	The reader can check every claim and see how old the data is

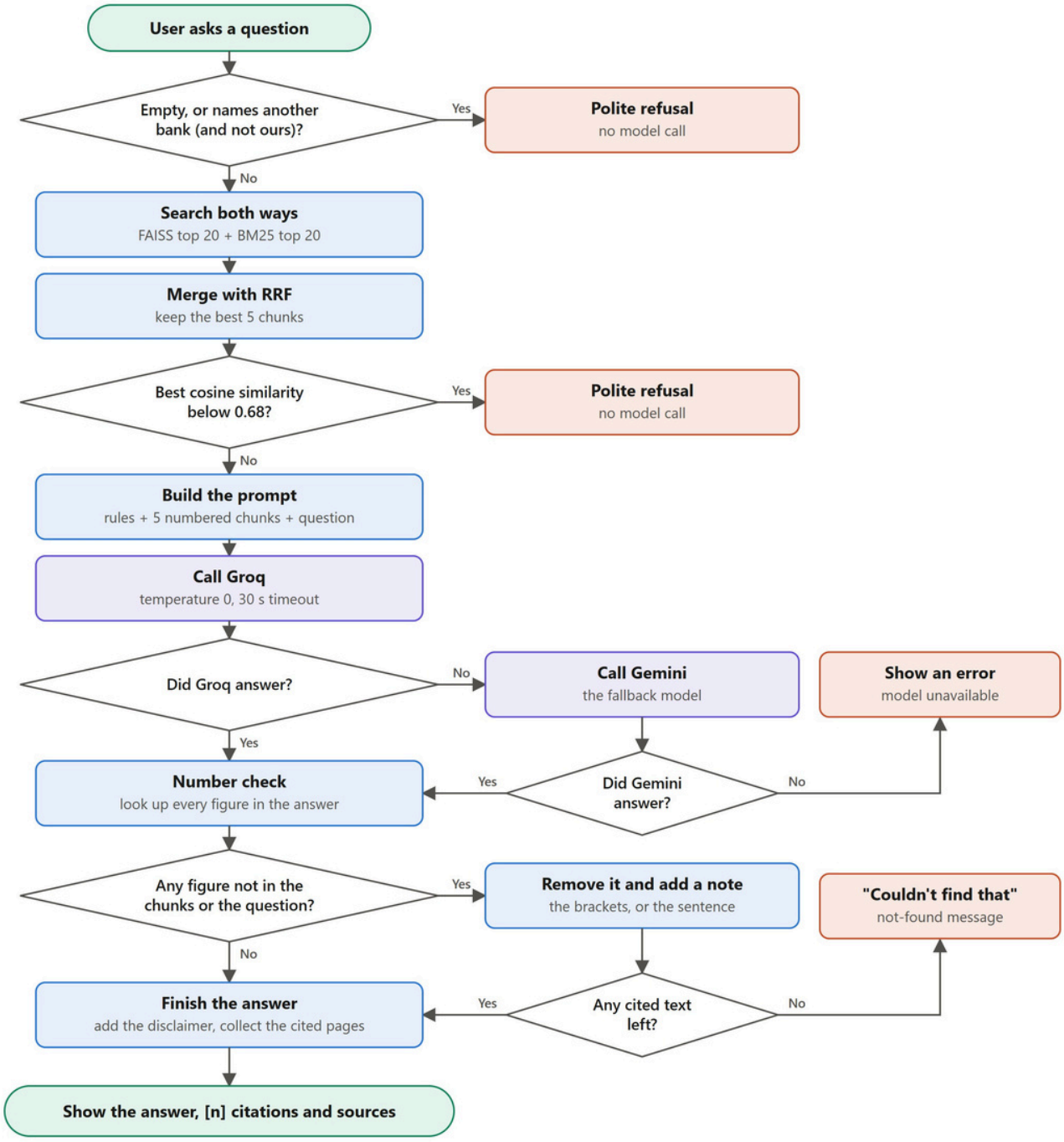


Figure 3. The path of one question through the pipeline.

5. Final Outcome

Theresultis aworking chat application over the bank's loan pages, tested on a written question set. The screenshots below are from the deliverables app, running on real data on 4 November 2026.

5.1 What was built

Item	Result
Pages in the knowledge base	86 web pages and 3 PDFs
Words / sections / tables	72,441 / 218 / 115
Page chrome removed	186,662 of 252,186 words on the raw web pages (74%)
Chunks	415, median 214 words, largest 450 words
PDFs left out	42: 14 scanned, 18 not English or garbled, 7 footer links, 2 blank forms, 1 wrong length
Automated tests	135 passed on 4 November 2026

5.2 Screenshots

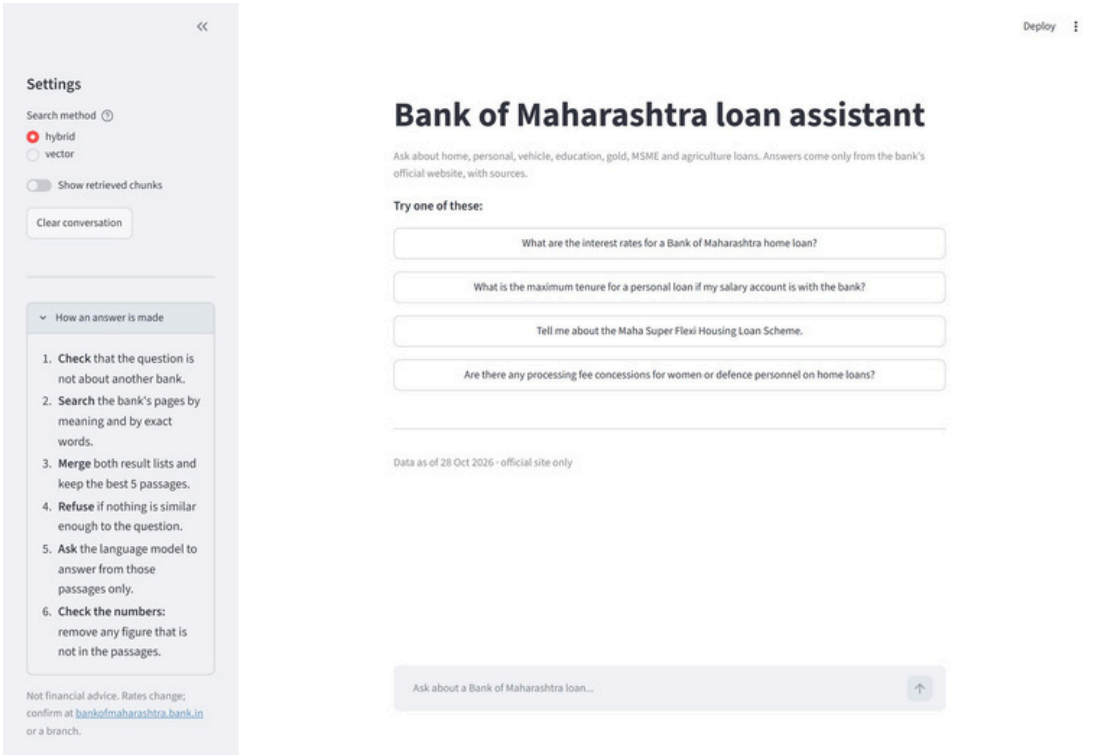


Figure 4. The home screen: example questions, search settings and a plain-language explanation.

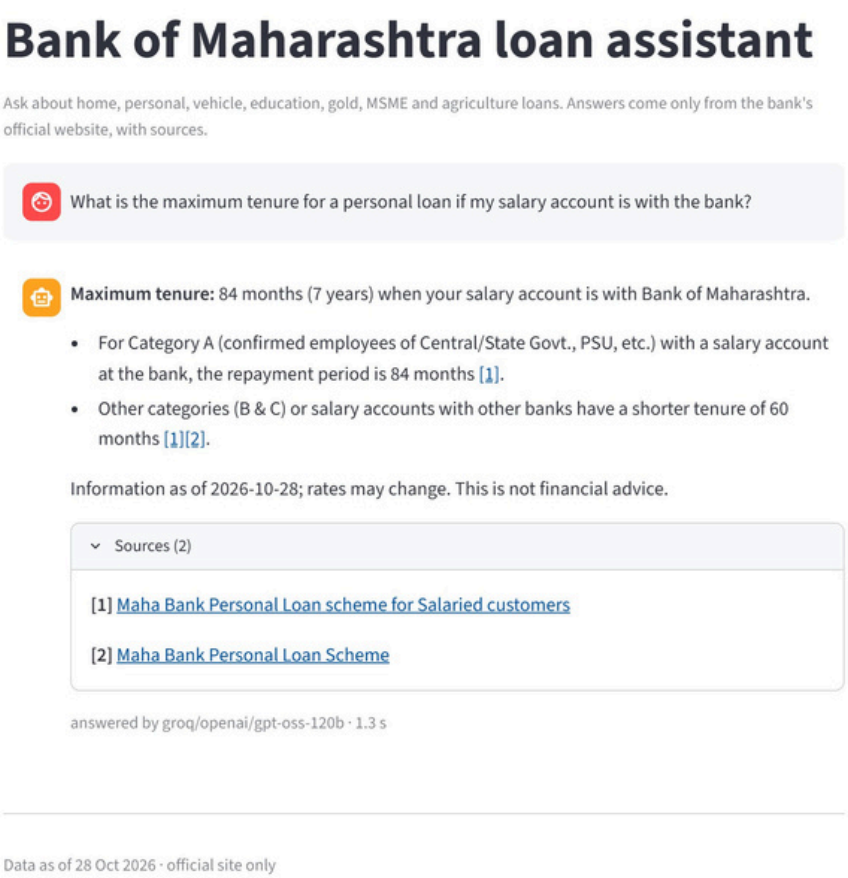


Figure 5. A real answer with linked citations, its sources, the model used and the time taken.

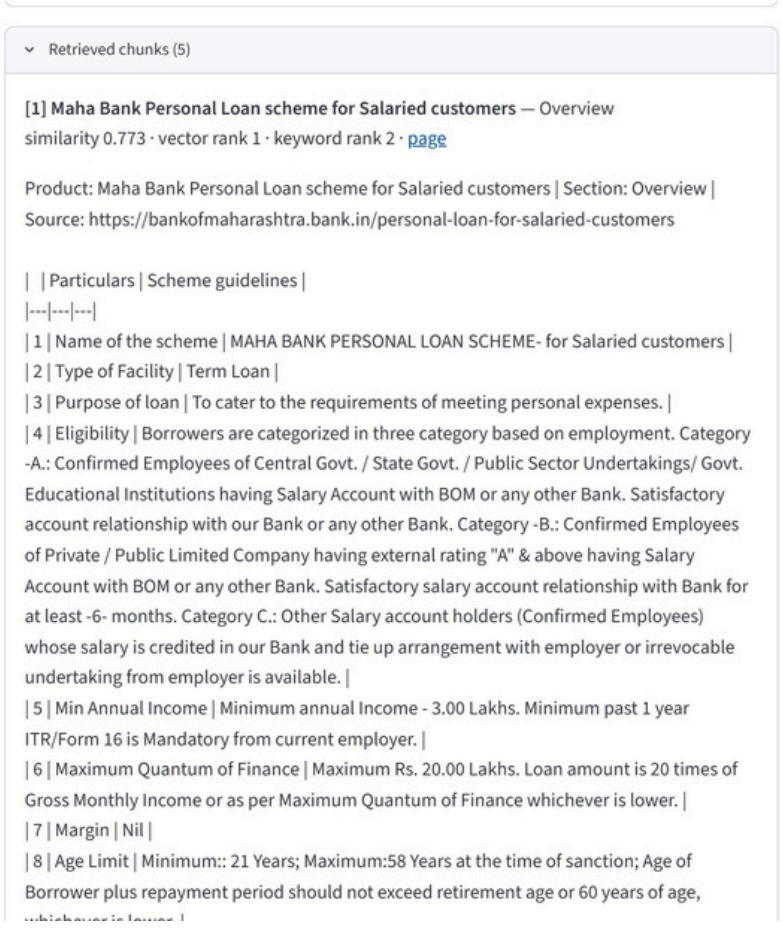


Figure 6. The chunk viewer shows exactly what the model was given, with the scores that chose each passage.

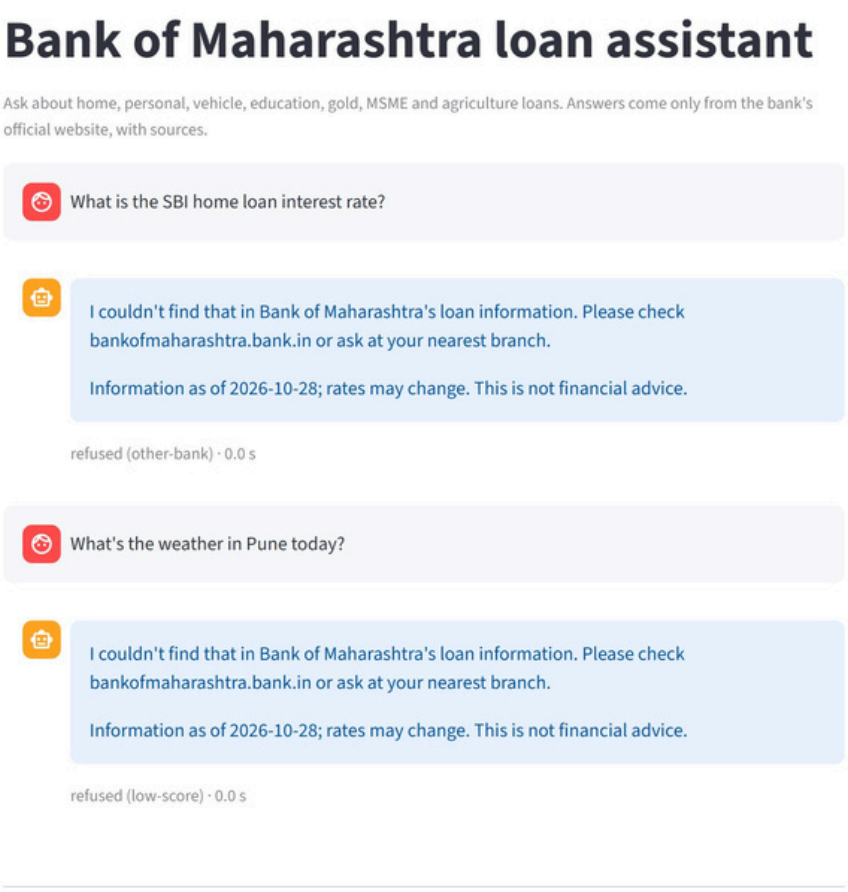


Figure 7. Two refusals: one for another bank and one for an off-topic question. Neither calls the model.

5.3 Sample outputs

Theseanswersarecopiedword for word from the saved evaluation run of 28 October 2026 (hybrid search), in eval/results/.

A correct, specific answer

Question: What is the maximum tenure for a personal loan if my salary account is with the bank?

Maximum tenure: 84 months

- For Category A salaried customers who have a salary account with Bank of Maharashtra, the personal loan can be repaid over **84 months**[1].
- This is also reflected in the FAQs, which list "Salary Account with Bank of Maharashtra – 84 months" for Category A[2].

Information as of 2026-10-28; rates may change. This is not financial advice.

Hand grade: 2 of 2. Sources: the personal loan scheme pages.

Keyword search finding the right page

Question: How much can I borrow against my fixed deposit and is there a processing fee?

You can borrow up to 90% of your Fixed Deposit's value, and there is no processing fee.

- Loan amount: up to **90% of the FD's value**[4]
- Processing fee: **NIL** (no processing fee)[4]

Information as of 2026-10-28; rates may change. This is not financial advice.

Hand grade: 2 of 2. Vector-only search missed the /lad page and declined to answer this question.

A refusal

Question: What's the weather in Pune today?

I couldn't find that in Bank of Maharashtra's loan information. Please check bankofmaharashtra.bank.in or ask at your nearest branch.

Information as of 2026-10-28; rates may change. This is not financial advice.

Hand grade: 2 of 2 (a refusal is the right outcome here). Best cosine similarity was 0.497, below the 0.68 limit, so no model was called.

A partly correct answer

Question: What are the interest rates for a Bank of Maharashtra home loan?

Base interest rate: 7.25% p.a. for a standard Bank of Maharashtra home loan[2]

Additional concession: Women borrowers receive a 0.05% interest-rate concession, i.e., the rate is 0.05% lower than the standard rate[1]

Key points

•Standard rate: **7.25% p.a.**

•Rate may vary based on credit score, loan amount, profession, etc., as noted in the FAQs[1]

Note: one figure in the first draft of this answer is not written on the bank's pages, so it was left out.

Information as of 2026-10-28; rates may change. This is not financial advice.

Hand grade: 1 of 2. It misses the CIBIL-based range. The number check removed an invented "7.20%" and left the note you see above.

5.4 Evaluation results

The 25test questions are22 about loans and 3 that should be refused. The expected facts were written by hand from the official pages. Both search modes were tested on 28 October 2026.

Metric	Vector only	Hybrid
Correct page in the top 5 (hit@5)	21/22 (95%)	22/22 (100%)
Correct page ranked first (hit@1)	17/22 (77%)	19/22 (86%)
All key facts in the answer (automatic)	16/22 (73%)	18/22 (82%)
Hand grade 2, correct and complete	**22/25 (88%)**	**24/25 (96%)**
Gemini judge grade 2	19/25 (76%)	20/25 (80%)
Out-of-scope questions refused	3/3	3/3
Real questions wrongly refused	1	0
Average answer time	2.17 s	0.89 s

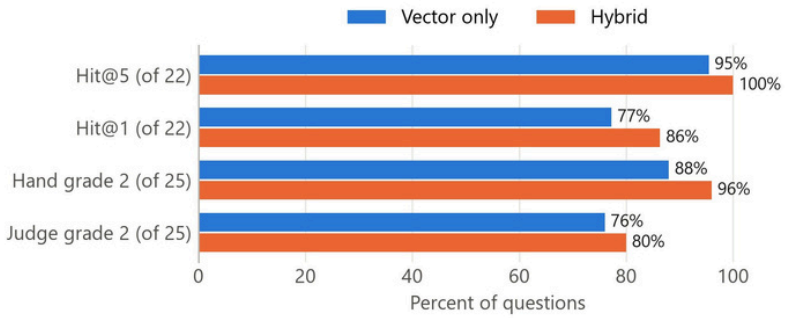


Figure 8. Vector-only and hybrid search compared. Both meet the 80% target.

How to read these numbers:

- With 25 questions, one question is worth 4 points. The hybrid gain rests mainly on one question (q13), so treat it as encouraging and not as proof.
- The Gemini judge is stricter than the hand grades because it expects every key fact, including ones the question did not ask for. It is a second opinion, not ground truth.
- The gap in answer time is mostly variation on the model's side. It does not mean hybrid search is faster.
- The walkthrough notebook re-measured retrieval live with its own code and matched the saved run exactly: hit@5 of 21/22 and 22/22, hit@1 of 17/22 and 19/22.

5.5 What did not work

Case	What happened	Status
q01: home-loan rate (both modes)	Gives 7.25% but misses the CIBIL-based range. The rate table is mostly numbers, which embed poorly, and the page says "Housing Loan" where people say "home loan"	Open. Four fixes were tried and dropped. The likely real fix is a reranker or a structured rate table
q01: invented "7.20%"	The model subtracted 0.05 from 7.25 despite a prompt rule against arithmetic	Fixed by the number check
q13: fixed deposit (vector)	The /lad page was not in the top 5, so the assistant said it could not find the answer	Fixed by hybrid search, which finds the page at rank 4
q10: education collateral (vector)	Mentioned only the Model scheme and missed the no-collateral Maha Scholar scheme	Graded correct in hybrid mode

5.6 Limitations and next steps

Limitations

- Answers come only from pages scraped on 28 October 2026, and some of those pages are themselves out of date.
- The number check catches a wrong figure, not a wrong statement in words. It only asks whether a figure appears somewhere in the retrieved text, and it does not check an invented whole number below 10 without a unit.
- Rate questions are the weakest area. 12 pages are thin after cleaning, and scanned PDFs are not read.
- English only, one question at a time with no conversation memory, and free-tier API limits apply.

Next steps

- Add a cross-encoder reranker and a structured rate table, to fix the q01 weakness.
- Add OCR for the 14 scanned PDFs, and a scheduled re-scrape with change detection.
- Add query rewriting ("home loan" to "housing loan"), conversation memory, and Hindi and Marathi.
- Build a larger test set with independent graders.

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